

Lecture 8 - Networks and Telecommunications

Overview

Networks and Telecommunications involve **computer networks**, **topologies**, and **protocols** that enable reliable communication and Internet services.

Topics

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| 1. Telecommunications | 6. Network Management |
| 2. A Computer Network | 7. Internet Protocols |
| 3. Types of Computer Networks | 8. Domain(s) |
| 4. Network Topologies | 9. Internet, Internet Service Providers, and the World Wide Web |
| 5. Network Protocols | 10. Search Engines |

Objectives / Outcome

1. Understand the fundamental concepts of **telecommunications** and **computer networks**.
2. Identify and describe different types of **computer networks** and their **topologies**.
3. Explain various **network protocols** and the importance of **network management**.
4. Understand key terms related to networking, including **Internet protocols**, **domains**, **ISPs**, and the **World Wide Web**.
5. Describe the function and importance of **search engines** in navigating the Internet.

Telecommunications

The transmission of **information** over significant distances through **electronic means**, including **telephone**, **radio**, **television**, and **computer networks**.

Importance

Telecommunications form the **backbone** of modern communications, enabling **data transmission** between **computers**, **mobile devices**, and other **electronic systems**.

Components

- **Transmitters:** Devices that convert **information** into **signals** for transmission.
- **Receivers:** Devices that convert transmitted **signals** back into usable **information**.
- **Communication Channels:** The medium through which **signals** are transmitted (e.g., **cables**, **wireless signals**).

Example

Common examples include **the Internet**, **mobile phone networks**, and **satellite communication systems**.

A Computer Network

Definition

Computer Network: A collection of interconnected devices (computers, servers, printers, etc.) that share resources and information.

Purpose

Facilitates communication and resource sharing between devices, enabling tasks like file sharing, internet access, and communication (email, instant messaging).

Basic Components

- **Nodes:** Devices connected to a network (e.g., computers, printers).
- **Links:** The physical or logical connections between nodes (e.g., cables, wireless signals).

Types of Network Configurations

- **Client-Server:** Centralized server provides resources and services to multiple clients.
- **Peer-to-Peer:** All devices share resources equally without a centralized server.

Types of Computer Networks

1. Local Area Network (LAN)

Definition: A network that connects **devices** within a limited area, such as a **home**, **school**, or **office building**.

Example: A **school network** connecting **computers** and **printers** within a classroom.

2. Wide Area Network (WAN)

Definition: A network that covers a large **geographic area**, often composed of multiple **LANs**.

Example: The **Internet** is the largest **WAN**, connecting networks worldwide.

3. Metropolitan Area Network (MAN)

Definition: A network that spans a **city** or a large **campus**.

Example: A **city-wide Wi-Fi network**.

4. Personal Area Network (PAN)

Definition: A small network for connecting **devices** close to an individual, typically within a range of a few meters.

Example: **Bluetooth** connection between a **smartphone** and a **wireless headset**.

Network Topologies

Definition

Network Topology: The arrangement or layout of different elements (**links**, **nodes**) in a **computer network**.

Common Topologies

- Bus Topology:** All **devices** share a single **communication line**.
 - Advantage:** Easy to install.
 - Disadvantage:** If the main cable fails, the entire network goes down.
- Star Topology:** All **devices** are connected to a **central hub**.
 - Advantage:** Easy to add or remove devices without affecting the network.
 - Disadvantage:** If the central hub fails, the entire network is affected.
- Ring Topology:** Devices are connected in a **circular fashion**.
 - Advantage:** Data flows in one direction, reducing the chance of packet collisions.
 - Disadvantage:** A failure in any cable or device disrupts the entire network.
- Mesh Topology:** Every **device** is connected to every other device.
 - Advantage:** Highly reliable, as there are multiple paths for data to travel.
 - Disadvantage:** Expensive and complex to implement.
- Hybrid Topology:** Combines two or more different types of topologies.
 - Advantage:** Flexible and scalable.
 - Disadvantage:** Can be complex to design and maintain.

Network Protocols

Definition

Network Protocol: A set of **rules** and **conventions** for **communication** between **network devices**.

Key Network Protocols

- Transmission Control Protocol/Internet Protocol (TCP/IP):**
 - Foundation of **Internet communication**.
 - Responsible for routing **data** between devices.
- Hypertext Transfer Protocol (HTTP/HTTPS):**
 - Protocol for transferring **web pages** on the **Internet**.
 - HTTPS** adds a layer of **security** through **encryption**.
- File Transfer Protocol (FTP):**
 - Used to transfer **files** between computers on a **network**.
- Simple Mail Transfer Protocol (SMTP):**
 - Protocol for sending **emails**.
- Domain Name System (DNS):**
 - Converts human-readable **domain names** (e.g., **www.example.com**) into **IP addresses**.

Network Management

Definition

Network Management: The process of **administering**, **managing**, and **operating** a **data network** using a **network management system**.

Functions of Network Management

1. **Monitoring:** Tracking the **performance** and **availability** of network resources.
2. **Configuration Management:** Managing the **setup** of network devices and systems.
3. **Fault Management:** Detecting, logging, and addressing **network problems**.
4. **Security Management:** Protecting the network from **unauthorized access** and **attacks**.
5. **Performance Management:** Ensuring network **efficiency** and **optimization**.

Tools

Network management software: Examples include **SolarWinds**, **Nagios**, and **PRTG**.

Terms and Definitions

- **Bandwidth:** The maximum rate of **data transfer** across a **network path**.
- **Latency:** The time it takes for a **data packet** to travel from the **source** to the **destination**.
- **Packet:** A **unit of data** transmitted over a **network**.
- **Router:** A device that forwards **data packets** between **computer networks**.
- **Switch:** A device that connects devices within a **LAN**, directing **data** to its correct **destination**.
- **Firewall:** A **security device** that monitors and controls incoming and outgoing **network traffic** based on predetermined **security rules**.

Internet Protocols

Definition

Internet Protocol (IP): The principal **communications protocol** in the Internet protocol suite for relaying **data** across networks.

Versions

- **IPv4:** Uses 32-bit addresses, allowing for approximately **4.3 billion** unique addresses.
- **IPv6:** Uses 128-bit addresses, providing a vastly larger **address space** to accommodate the growing number of **devices** connected to the Internet.

Key Functions

- **Addressing:** Assigning **unique addresses** to devices on a network.
- **Routing:** Determining the best **path** for data to travel across interconnected networks.

Domain(s)

Definition

Domain: A unique name that identifies a **website** on the **Internet** (e.g., **www.example.com**).

Components

- **Top-Level Domain (TLD):** The last segment of the domain name (e.g., **.com**, **.org**, **.edu**).
- **Second-Level Domain (SLD):** The part of the domain that precedes the TLD (e.g., **"example"** in **www.example.com**).

Domain Name System (DNS)

The system that translates **domain names** into **IP addresses**, allowing **browsers** to load Internet resources.

Internet, Internet Service Providers, and the World Wide Web

Definition

Internet: A global network of interconnected computer networks that use the TCP/IP protocol to communicate.

Internet Service Providers (ISPs): Companies that provide access to the Internet.

World Wide Web (WWW): A system of interlinked hypertext documents and multimedia accessible via the Internet.

Function

- **Internet:** Enables communication and access to information, services, and resources worldwide.
- **ISPs:** Provide Internet access, email services, web hosting, etc.
- **WWW:** Uses web browsers to retrieve and display content from web servers via HTTP/HTTPS.

Examples

- **ISPs:** Comcast, AT&T, Verizon.

Search Engines

Definition

Search Engine: A software system designed to search for **information** on the **World Wide Web** and present results based on **user queries**.

Function

- Indexes vast amounts of **web content**.
- Uses **algorithms** to determine the relevance of each result.

Popular Search Engines

- **Google:** The most widely used search engine.
- **Bing:** Microsoft's search engine.
- **DuckDuckGo:** A privacy-focused search engine.

Importance

- Facilitates easy access to **information** on the web.
- Helps users find **relevant content** quickly.

Summary

This lecture covers key concepts in **networks** and **telecommunications**, including **telecommunications**, **types of computer networks**, **network topologies**, **network protocols**, and the basics of **Internet protocols**, **domains**, **ISPs**, the **WWW**, and search engines.